

Date	03 July 2026
Team ID	SWTID-2026-4799
Team Size	5
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Project Name	Machine Learning—Based Credit Card Approval Prediction System
Maximum Marks	4 Marks

Define Problem Statements

1. Problem Statement

Financial institutions receive thousands Of credit card applications every day. The traditional approval process mainly depends on manual verification of applicant information such as income, employment status, credit history, financial background, and Other eligibility factors. Manual evaluation requires considerable time and human effort, Often resulting in delayed decisions, inconsistent assessments, and increased operational costs. There is a need for an intelligent and automated solution that can accurately predict credit card approval using Machine Learning techniques while supporting faster and more reliable decision-making.

2. Existing System

Aspect	Description
Application Evaluation	Manual verifvation by banking staff
Decision Making	Human judgement
Processing Time	Several hours or days
Consistency	Varies between evaluators
Scalability	Limited for large numbers Of applications
Operational Cost	High
Risk of Human Error	High
Automation	

3. Problems Identified in the Existing System

- Time-consuming verification process.
- Heavy dependency on manual decision-making.
- Inconsistent approval decisions.
- Increased operational costs.
- Difficulty processing large application volumes.
- Higher probability of human error.
- Limited scalability.

- Slow customer response.

4. Root Cause Analysis

Problem	Root Cause
Slow processing	Manual verification Of applications
Inconsistent decisions	Human judgement varies
High operational cost	Large workforce involvement
Customer dissatisfaction	Long waiting time
Poor scalability	Increasing number of applications

Note

The identified problems highlight the need for an intelligent, automated, and data-driven credit card approval prediction system capable of improving efficiency, consistency, and decision accuracy.

5. Proposed Solution

The proposed Machine Learning—Based Credit Card Approval Prediction System automates the evaluation of credit card applications using historical applicant data and trained Machine Learning models. The system analyses applicant information, predicts whether the application is likely to be approved or rejected, and provides rapid, consistent, and reliable decision support for financial institutions.

6. Project Objectives

1. Automate the credit card approval prediction process.
2. Reduce manual verification effort.
3. Improve prediction accuracy using Machine Learning.
4. Minimize processing time.
5. Assist financial institutions in decision-making.
6. Improve customer experience.
7. Generate reliable prediction reports.
8. Build a scalable web-based prediction platform.

7. Project Scope

The scope of this project includes applicant registration, secure authentication, credit card application submission, data preprocessing, Machine Learning prediction, approval result generation, database management, prediction history, administrative reporting, and analytics. The system focuses on predicting credit card approval based on applicant information while providing an intuitive web interface for both applicants and administrators.

8. Expected Benefits

Benefit	Description
Fast Prediction	Provides approval results within seconds.
Higher Accuracy	Uses Machine Learning algorithms for intelligent prediction.

Reduced Cost	Minimizes manual processing effort.
Consistency	Produces standardized prediction results.
Improved Customer Experience	Faster application processing.
Scalability	Supports increasing numbers of applications.
Better Decision Support	Assists administrators with data-driven decisions.

9. Conclusion

Conclusion

The Machine Learning—Based Credit Card Approval Prediction System addresses the limitations of traditional credit card approval processes by introducing intelligent automation, faster prediction, improved accuracy, enhanced scalability, and reliable decision support. The proposed solution contributes to more efficient banking operations while improving the overall customer experience.